

SASTRA DEEMED UNIVERSITY

(A University under section 3 of the UGC Act, 1956)

End Semester Examinations

Naveen

May 2024

Course Code: **EEE101**

Course: **BASIC ELECTRICAL ENGINEERING**

QP No. : **U039-2**

Duration: 3 hours

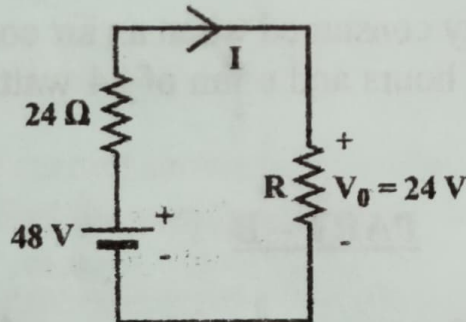
Max. Marks: 100

PART - A

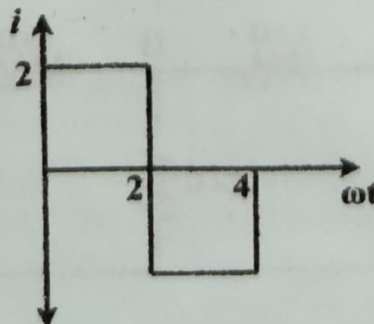
Answer all the questions

10 x 2 = 20 Marks

1. In the given circuit, find the value of current (I) and resistance (R).



2. A sinusoidal alternating current i is given by the expression $i = 24 \sin 314 t$. find its time period.
3. Determine the average value of the waveform shown below.



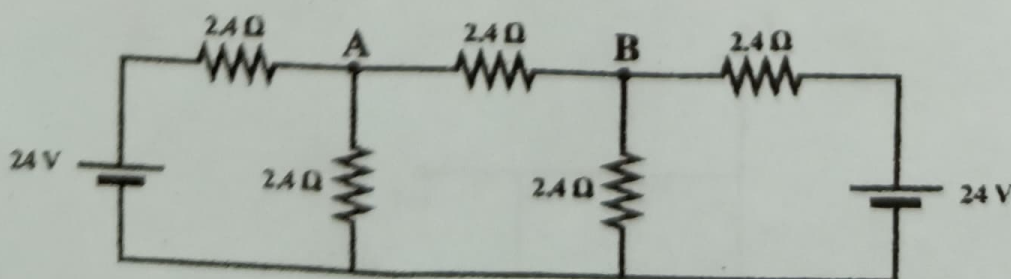
4. Write and state the rule that helps to find the direction of force experienced by a current carrying conductor placed in a magnetic field.
5. Mention the function of commutator in DC generator and DC motor.
6. What would happen to the operation of a DC motor if the motor is started without a starter?
7. Identify the type of induced emf in single phase transformer and DC generator. Justify your answer.
8. A coil has 2400 turns. A current of 24 A causes a flux of 24 milli weber to link the coil. Determine the self inductance of the coil.
9. Mention the types of wiring that can be employed for domestic electrical connections.
10. Calculate the total energy consumed when an air conditioner of 2400 watts is operated for 24 hours and a fan of 24 watts operated for 24 hours?

PART – B

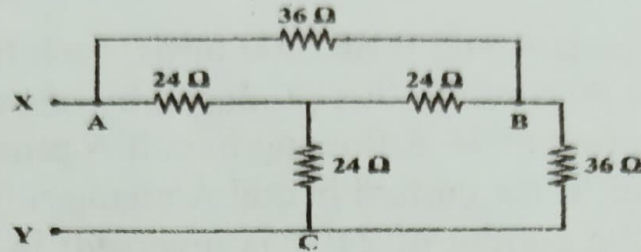
Answer any Four questions

4 x 15 = 60 Marks

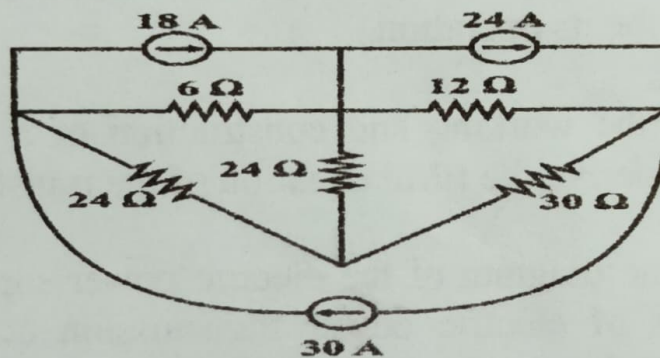
11. a) Determine the current through all the resistors using mesh analysis. Assume a lamp is connected between points A and B whose resistance is 2.4 ohms. State the illumination condition of the lamp. (10)



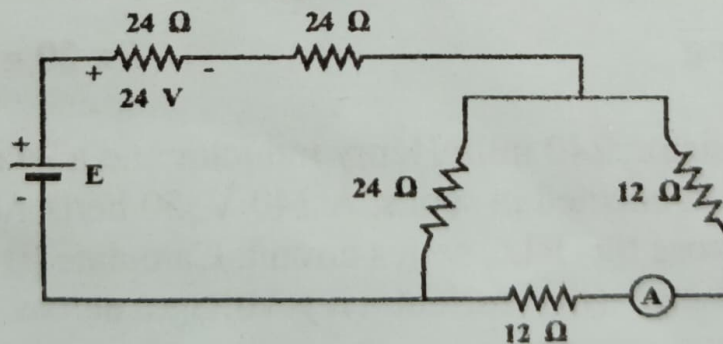
- b) Find the equivalent resistance between X and Y. A 24 V ideal voltage source is connected across X and Y. Determine the current supplied by the source. (5)



12. a) Write the node equations for the circuit shown in the figure and determine the node voltages. Also, determine the current through the 6 ohms resistor. (10)



- b) Determine the current shown by the ammeter for the circuit shown below. Also, find the supply voltage. (5)



13. a) An iron ring has a mean diameter of 24 cm. The relative permeability of the material is 2400. A coil with 2400 turns is wound uniformly over the ring and is supplied with a current of 2.4 A. Determine the magnetomotive force, magnetic field intensity and magnetic flux density. (5)

b) Draw the equivalent circuit with its voltage and current equation for a DC shunt, and DC series motor. (5)

c) Two identical coils A and B of 2400 turns, each lie in a parallel plane such that 90% of the flux produced by one coil links with the other. A current of 24 A flowing in coil A produces a flux of 0.24 milli weber. If the current in coil A changes from + 24 A to - 24 A (or if the current of 24 A is reversed) in 0.24 seconds, calculate the mutual inductance and the EMF induced in coil B. (5)

14. Explain, with a neat diagram, the constructional features of a DC machine and describe its operation.

15. Discuss in detail, the working and construction of a single-phase transformer. Also, derive the EMF equation of the transformer.

16. Draw the single line diagram of the electric power supply scheme. List out the types of electric power transmission & distribution systems and explain the distribution of power from generating station to consumers.

PART - C

Answer the following

1 x 20 = 20 Marks

17. a) A 24-ohm resistor, 240 milli Henry inductor and a 24 micro farad capacitor are connected in series. A 240 V, 50 hertz AC supply is connected across this RLC series circuit. Calculate (i) impedance (ii) power factor (iii) current (iv) voltage across resistance, inductance, and capacitance, (v) complex power supplied by the source and (vi) draw its phasor diagram. (10)

b) State the importance of earthing in domestic wiring. With the help of a neat diagram, explain the arrangement of pipe earthing. (10)
